

Unit 11 Machine Learning Assignment
SPOTIFY SONG RECOMMENDER SYSTEM PRESENTATION TRANSCRIPT

Introduction

Hello and welcome to this presentation where I will walk through the build of a content-based song recommender system for Spotify. This presentation will cover the: dataset selected, exploratory data analysis (EDA), data preprocessing, and feature engineering conducted, design of the two machine learning (ML) models developed - one classical ML and one deep learning (DL) model, evaluation of the models using metrics and the validation strategy, explainability of the models using SHAP, bias analysis, ethical considerations, deployment considerations and caveats, and a live demonstration of the model.

Brief background of Spotify & The Dataset

Spotify is a music, podcast, and audiobook streaming platform with over 760 million active monthly listeners globally. It hosts over 100 million songs, 7 million podcast titles, and 700k audiobooks (Spotify, no date). Machine Learning (ML) is embedded into the platform to recommend songs and other materials to users based on their listening data, and that is what will be the focus.

The dataset used is a publicly available one from Kaggle, 'Spotify Tracks'. It contains the metadata of 114,000 songs with varying genres and audio features such as loudness, energy, tempo, and speechiness. This doesn't include user interaction data, such as listening history or a user's song ratings, so a traditional recommender system based on user ratings cannot be developed. A content-based system was created along with simulated user taste profiles, defined by a preferred genre and

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specific audio feature. This means that a meaningful relevance signal can be constructed without real interaction data.

EDA, Data Preprocessing & Feature Engineering

EDA was conducted to understand the characteristics of the data and its variables, or features, to inform specific data preprocessing and feature engineering steps. This is a key step before model development, as this prevents misleading insights and low accuracy caused by patterns drawn from inconsistencies (Koukaras and Tjortjis, 2025). Figure 1 is a series of histograms displaying the distribution of numeric variables. Some notable findings: popularity is right skewed which indicates that most tracks are low popularity, or 'niche'. It has a mean of 33.28 which confirms that most songs are niche. Figure 2 is comprised of histograms solely for the audio features. 'acousticness', 'speechiness', 'liveness' are also right skewed, while 'loudness' and 'energy' are left skewed. Noting this as this informs a preprocessing step that was subsequently conducted. Figures 3 & 4 show the top and bottom 20 genres by track count, indicating that there is an even spread of songs across all genres. Figures 5, 6, & 7 show bivariate analyses. 'energy' and 'loudness' are strongly positively correlated with a coefficient of 0.76, which you can see visualised in figure 6, while 'acousticness' and 'energy' are strongly negatively correlated with a coefficient of -0.74. The median popularity across the top 15 genres doesn't vary drastically either, so it is unlikely that a song's popularity is greatly affected by its genre.

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Preprocessing of the data involved removing duplicate tracks, dropping 3 rows that were missing key fields like 'track_name' and 'genre' since it's tricky to impute them, and removing invalid entries like tracks with no duration. To combat the skewness present, those features were transformed using Yeo-Johnson power transform. This was used since applying min-max scaling would only resolve the range of the feature and not its shape. It's also crucial as while certain classical ML models, like RFs are more resilient to skewed data, skewed inputs can warp gradient-based learning in neural networks (Japkowicz and Stephen, 2002). Figure 8 shows the histograms post-transformation, with more normal distributions particularly for 'energy', 'speechiness' and 'liveness'.

On top of the raw metadata, several engineers were featured including: a genre-frequency feature to separate how popular a specific track is from how mainstream it is, interaction terms for energy times loudness and danceability times valence (positivity), two profile-track interaction features: 'genre_match' and 'mood_match', so that both models can access if a track matches the user's stated genre & mood preference as opposed to inferring the relationship from the raw numbers. and; a mood quadrant, based on Russell's circumplex model from music psychology, shown in figure 9 which places each track into: Happy and Excited, Calm and Content, Tense and Angry, or Sad and Depressed. This is based on its energy and valence.

ML Model Design

Two models were developed: a classical ML model, and a deep learning model.

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Using scikit-learn, the classical ML model chosen was Random Forest Regression. Random Forest, or RF, allows for nonlinear interactions in data, such as the ones between energy and loudness, and can adapt to high dimensional datasets such as this one, confirmed in Borup *et al's* (2023) paper on the model. In the model, multiple decision trees were built on samples of the engineered features (raw audio features, interaction-mood featured and profile-target features) with hyperparameters, such as the number of trees – 200, max depth, and minimum samples per leaf. Their predictions were averaged to produce a single relevance score for each track-user pair. This naturally reduces variance without needing to scale the features.

The DL model chosen was a Multilayer Perceptron (MLP), a type of feed-forward neural network, developed using TensorFlow. As defined by Sazlı (2006), feed-forward neural networks are a type of ANN where the data flows in one direction only. In this context, each track's audio features and a user's preferences are entered as input, passed through two hidden layers where the network learns combinations and interactions between them, before coming out as a single predicted relevance score between 0 and 1 in the output layer. The two hidden layers were of 64 and 32 units, with batch normalisation and a 20% dropout after each layer to combat overfitting since it prevents the network from relying on any single feature or pattern. The Adam optimiser with a learning rate of 0.001 was used and training halted early once validation stopped improving with a patience of 8 epochs (which is number of passes through the network).

Validation Strategy & Evaluation Metrics

As the dataset doesn't feature real user interaction data or profiles, robust validation was conducted to ascertain how likely it is that an outcome is due to chance (Gorriz *et al.*, 2024). To evaluate, a five-fold GroupKFold cross-validation was employed, in line with standard model selection methodology as described by Kohavi (1995) in his comparison of validation methods. This tests how well the models generalises to new users rather than ones it's already seen. It splits by the synthetic user profiles into 5 groups with an 80/20 train-test split, to ensure that no single user profile appears in both the training and test data and provides a more reliable estimate of real-world performance. Without this, data leakage could occur since the models could resort to memorising a user's preference from training rows and generalise that to unseen profiles, inflating performance.

Two evaluation metrics were used: root mean squared error (RMSE) and normalised discounted cumulative gain (NDCG). RMSE measures how far off the models' predicted relevance score is from the true relevance score on average across all track-user pairs. It is a good metric to use in this case since after transformation, the features possess a more normal distribution (Hodson, 2022). NDCG measures how well the models' ranked recommendations, rewarding them for placing the most relevant tracks near the top and penalising them when relevant tracks are placed lower down. This is similar to how Kshirsagar *et al* (2025) used the metric to evaluate their course recommendation ML model.

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Across the 5 folds, RF achieved a mean RMSE 0.031 with a standard deviation of 0.0006, compared to the MLP's mean of 0.034 with a standard deviation of 0.0009. The RF model is more precise at predicting relevance score of songs than the MLP, although this is marginal. Comparing the two NDCG means for each model of 0.969 for each, indicates that they both perform equally as well when ranking recommended songs.

Explainability with SHAP

Shapely Additive Explanations (SHAP) is an explainability technique that assigns an importance value to each feature for a particular prediction and works out how much each feature influenced the relevance score up or down compared to the average prediction (Lundberg and Lee, 2017). Figures 9 and 10 are summary plots of SHAP for each model. In both, you can see that 'genre_match' is the most important feature for both – high genre-match values push predictions upwards, while low values push them downwards, confirming that both models have learned that the most significant driver of relevance is genre match. However, RF learns more on raw audio features – 'instrumentalness', 'loudness', and 'acousticness', while the MLP gives more weight to the interaction and mood features that were engineered during preprocessing, such as the dance-valence interaction and mood quadrant dummy variables.

Bias & Diagnostic Analysis

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Disparities in performance across subgroups were tested, and this is a method of checking whether a content-based recommender like this under serves users with low activity. Divided by track popularity, RF performed on low-popularity songs versus high-popularity tracks, with a RMSE score of 0.39 vs 0.31 respectively, showing a bias towards more mainstream songs users may already listen to. The opposite was true MLP model, with a bias towards niche songs as seen by the scores of 0.0372 for high popularity and 0.0316 for low popularity. Regarding genre rarity, RF's RMSE was 0.0352 for rare genres and 0.0349 for common genres, indicating little bias. Conversely, MLP's RMSE for rare genres was 0.0235, indicating that while RF's RMSE is better on average overall, the MLP treats niche songs fairer.

Ethics

As with any ML model, there are key ethical considerations that must be accounted for.

Since the dataset is publicly available metadata and doesn't contain any personal user data or listening history, it carries little privacy risk in this context. However, a real recommender system in production would require strict adherence to data retention policies under GDPR, and explicit consent from users would be required since their sensitive behavioural data would be collected. Non-compliance with these data protection principles would result in high penalties and reputational damage (*Art. 83 GDPR - General conditions for imposing administrative fines*, 2018).

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Fairness is a key consideration. A false negative result, which in this context would be a niche track that aligns with the user's preferences not being recommended, would suppress the exposure and streams that smaller, independently managed artists get. Over time, this would compound into a significant amount of under-exposure and lost revenue from streaming royalties. A false positive result wouldn't be as problematic as the user would skip any songs they deem irrelevant.

Considering this, the MLP model would be favoured to mitigate this given its RMSE score for rare genres. If the RF was to be implemented, then consistent monitoring of its subgroup performance gap.

Deployment Considerations

For deployment into production, the trained model placed behind a REST API, wrapping a lightweight inference server like a managed endpoint on Google Vertex AI, rather than just exposing the raw one. Spotify has utilised the Google Cloud Platform since 2016, so opting for the Vertex AI would ensure seamless integration (*Spotify Cloud Infrastructure Powers Global Music Streaming*, no date). A REST API is best since each recommendation request is stateless, and each request only needs a user profile sent to receive ranked songs. It is also easily scalable which aligns with Spotify's desire for growth, and each request can be logged to monitor fairness.

Across an entire catalogue of songs (since the platform hosts over 100 million songs), scoring of every track against every user profile wouldn't work. A workaround is to use an approximate nearest-neighbour index, like Facebook AI Similarity

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Search (FAISS) (Gollapudi *et al*, 2023). Similar songs could be found across millions of vectors without comparing each one, by building a search index that immediately shortlists any close matches.

The RMSE and NDCG scores of the model should be monitored closely alongside fairness metrics, and retraining should occur on a rolling basis.

Demonstration

This is a recorded demonstration of the recommender system using the MLP model where the user's preferences can be entered to receive a list of songs which best match those requirements.

In this demonstration of the recommender system, I will run the demo twice and use two different genre examples. If I run this block of code, it prompts me to input my preferred genre, which I will do (afrobeat), pressing enter to confirm, and then it will ask me to confirm the mood. I'll input 'happy'. If we scroll down to the output, it has produced a list of tracks which best meets my preferences, ranked by the predicted relevance. If we scroll back up and I run the code again, I'll input 'pop' this time and I will input a mood of 'chill'.

The model has run and once again another list ranked by their relevance. That is the demonstration of the recommender system based on the preferences that a user has inputted.

Conclusions

To summarise, the content-based recommender system using the RF model is a better choice if raw predictive accuracy and consistence across folds is the main priority. However, the MLP produces fairer recommendations across both popularity and genre-rarity subgroups, which makes it the recommended choice for deployment in the real world, so long as its slightly higher variance across folds is monitored closely once in production.

Thank you for listening.

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